Facultatea de Automatică și Calculatoare

Universitatea Politehnica din Timișoara

AIRNEST – SMART IOT INDOOR AIR MONITORING STATION

Bachelor's thesis

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INTRODUCTION

The project presented today shows a smart indoor air quality station product. It is mostly destined for residential use and it lets the user easily visualize the data. This project meets the following requirements:

- Real-time reading and display of sensor read values
- Real-time storage of read data in Cloud
- Retrieval and processing of data in Cloud
- Data visualization through the website and mobile app
- Predictions generated based on gathered and existing data
- Download of raw data and mobile app

SIMILAR WORKS

Feature	Awair Element	uRADMonitor A3	Current Project
Hardware platform	Proprietary	Proprietary	Raspberry Pi 4
Monitored parameters	Temp, humidity, CO_2, TVOC, PM2.5	PM, O_3, HCHO, and others	Temp, humidity, CO, LPG, alcohol, H_2, air quality, sound
Web interface	No	Online platform	Yes
Mobile interface	Yes	No	Yes
Data export (CSV)	No	Limited	Yes
ML prediction	No	No	Yes
Open / customizable	No	Partial	Yes
Cost	~$299	~$350	Below $100

PROBLEM AND SOLUTION

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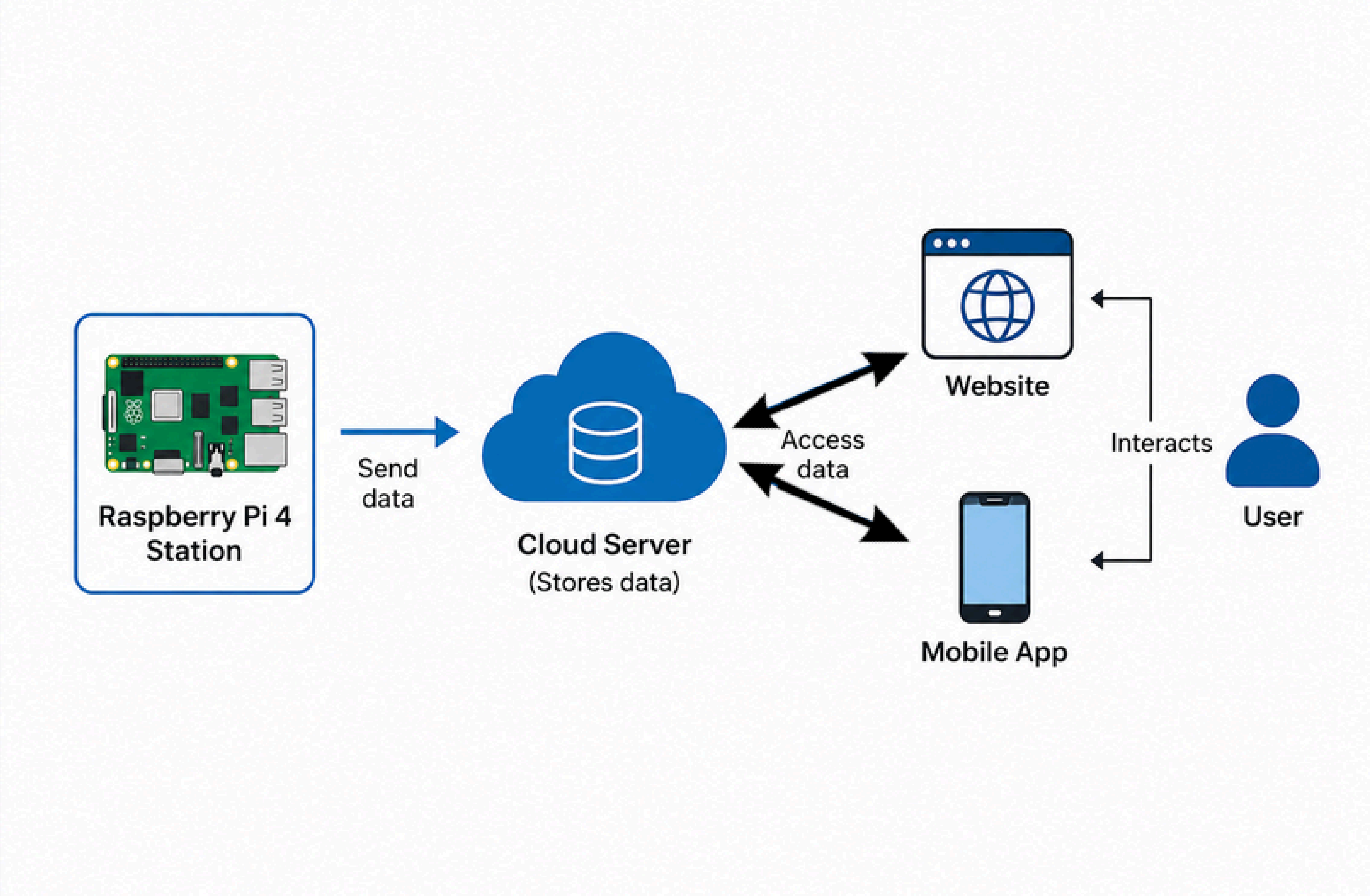
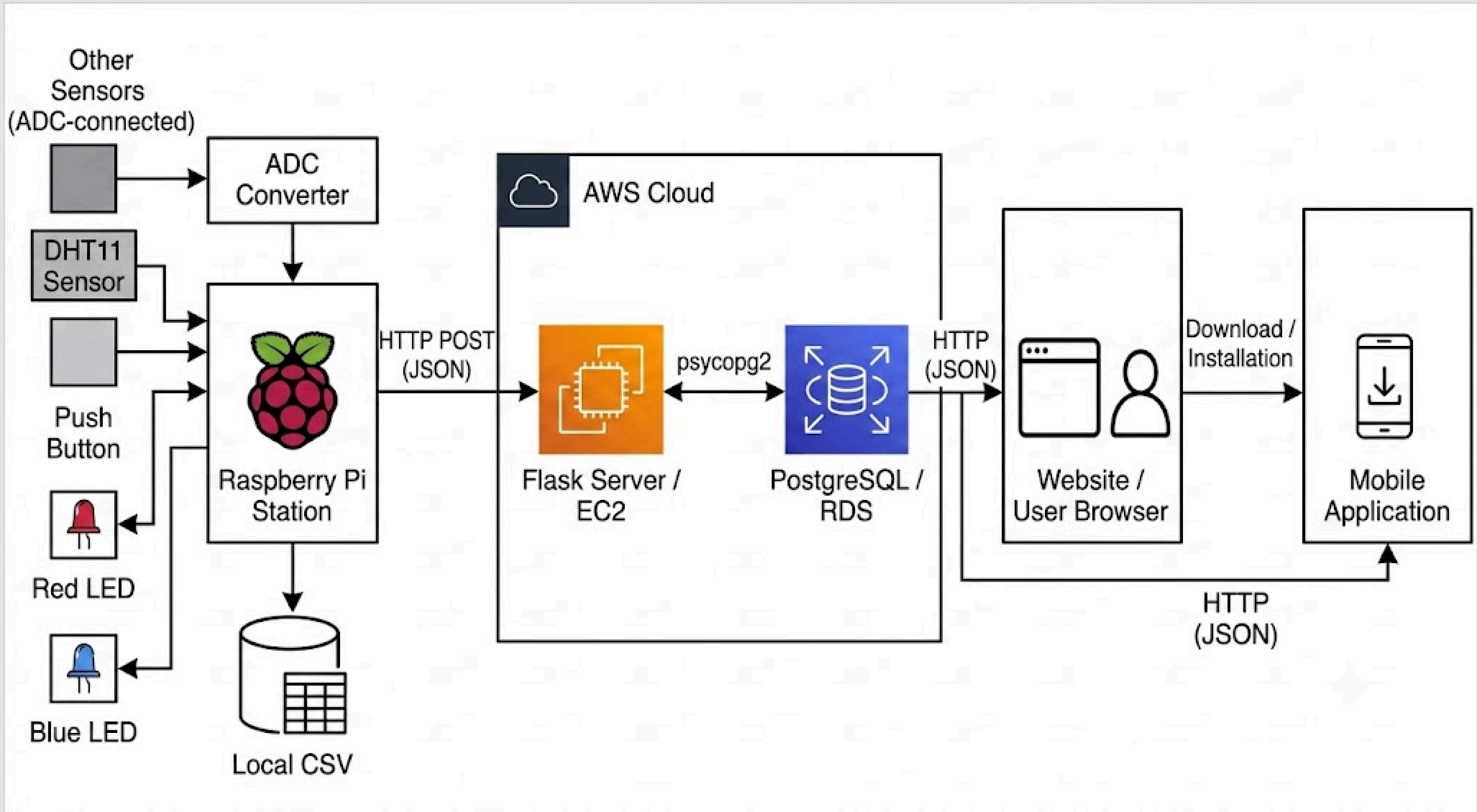
Problem: lack of affordable product options for indoor air quality monitoring.

Solution=> a product made after this project, which can be arranged in 3 parts:

- Hardware part, having a Raspberry Pi as base
- Website and mobile application (runs on android)
- Cloud storage for data (AWS are used)



SYSTEM ARCHITECTURE



HARDWARE ARCHITECTURE

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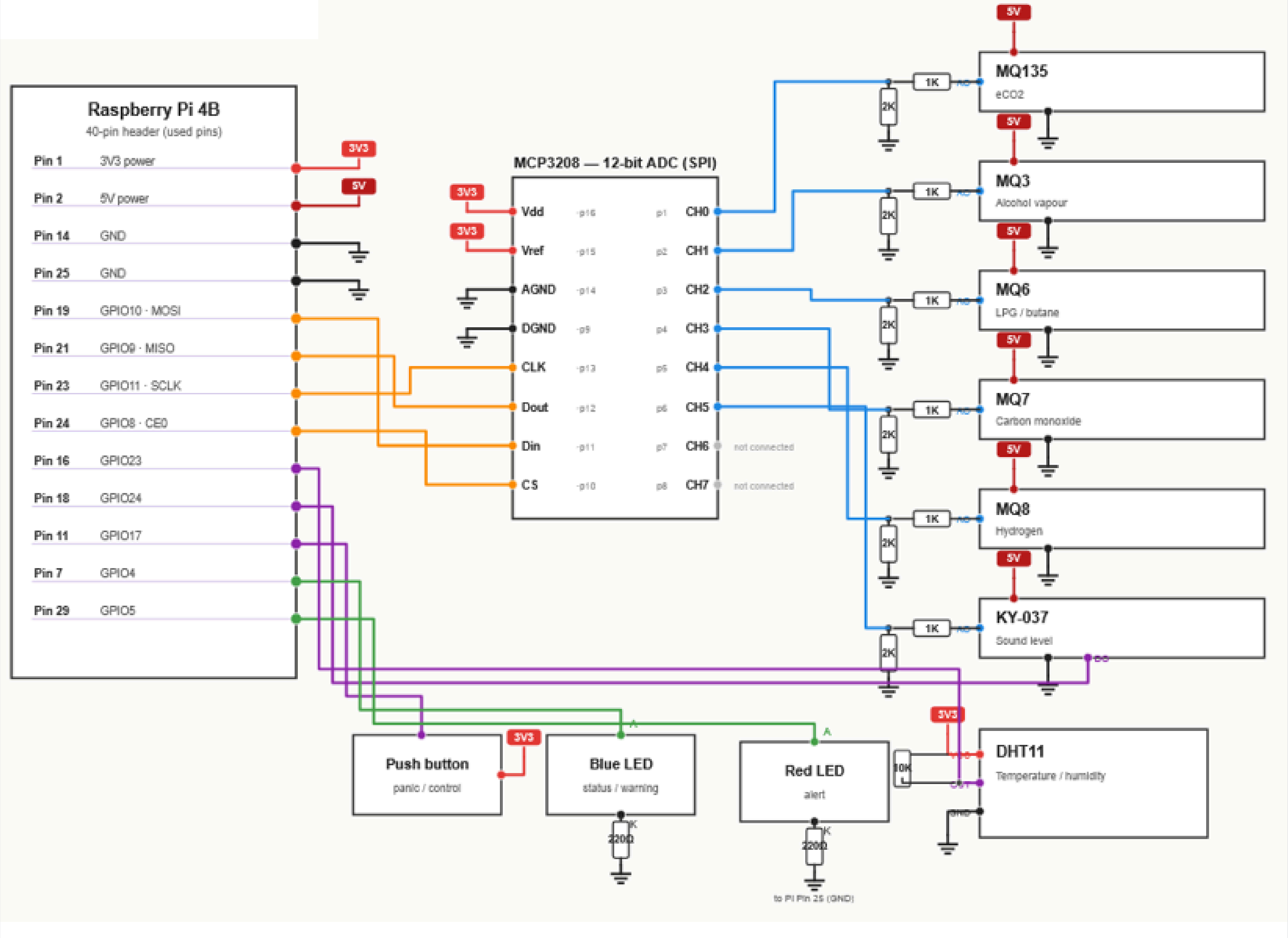
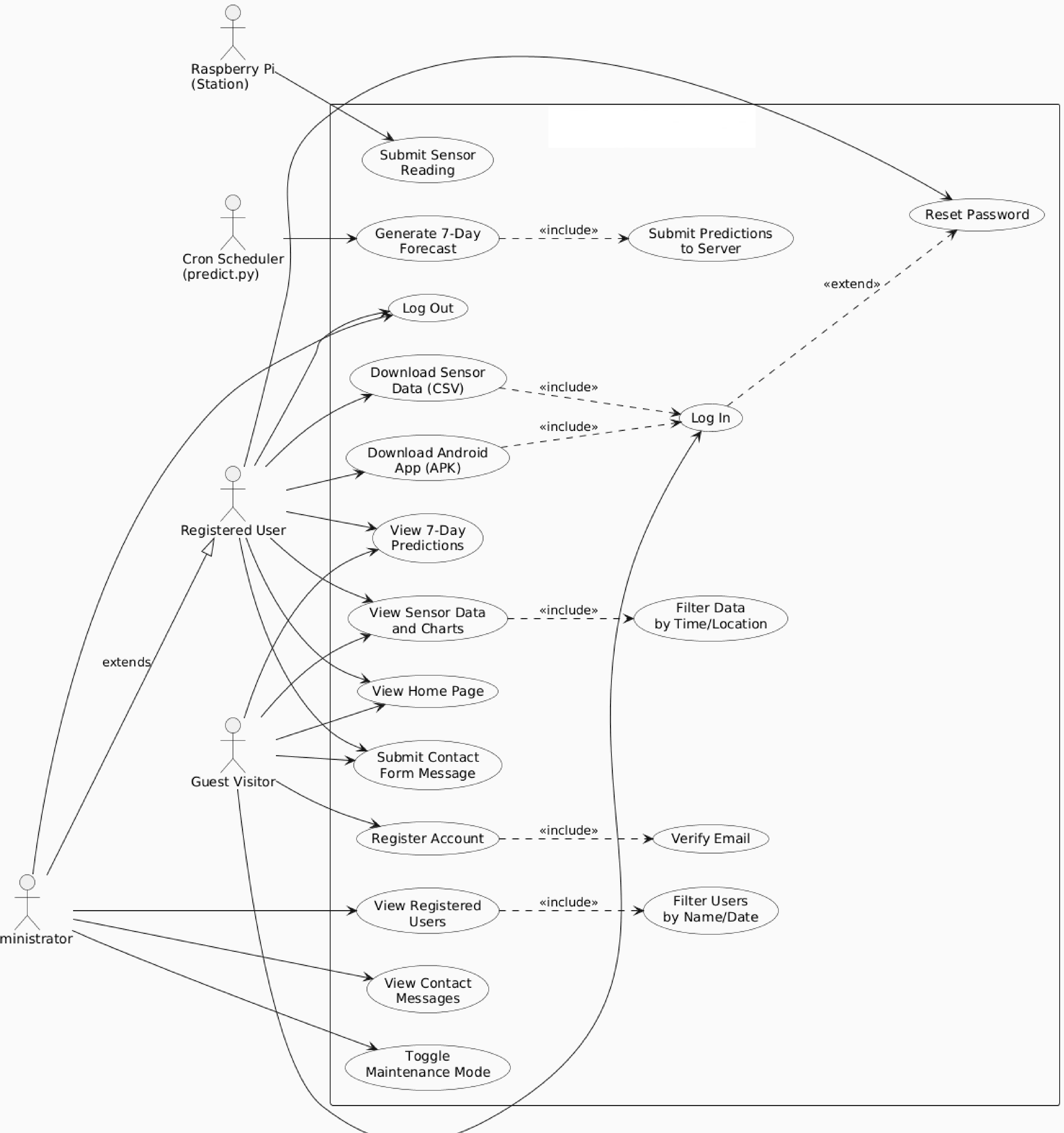




SOFTWARE ARCHITECTURE

MACHINE LEARNING

There are two models used:

- Prophet (used for predicting temperature and humidity up to 7 days ahead)
- Random Forest Classifier(used to classify if a future day will have dangerous air quality levels)

TESTING

To make sure the system is functioning properly, testing was conducted in two parts:

Data gathered and computed by the station to ensure its accuracy.
The generation of predictions and sending data.

Testing the website and mobile application so there are no errors and data is available to see when used by the user.

RESULTS

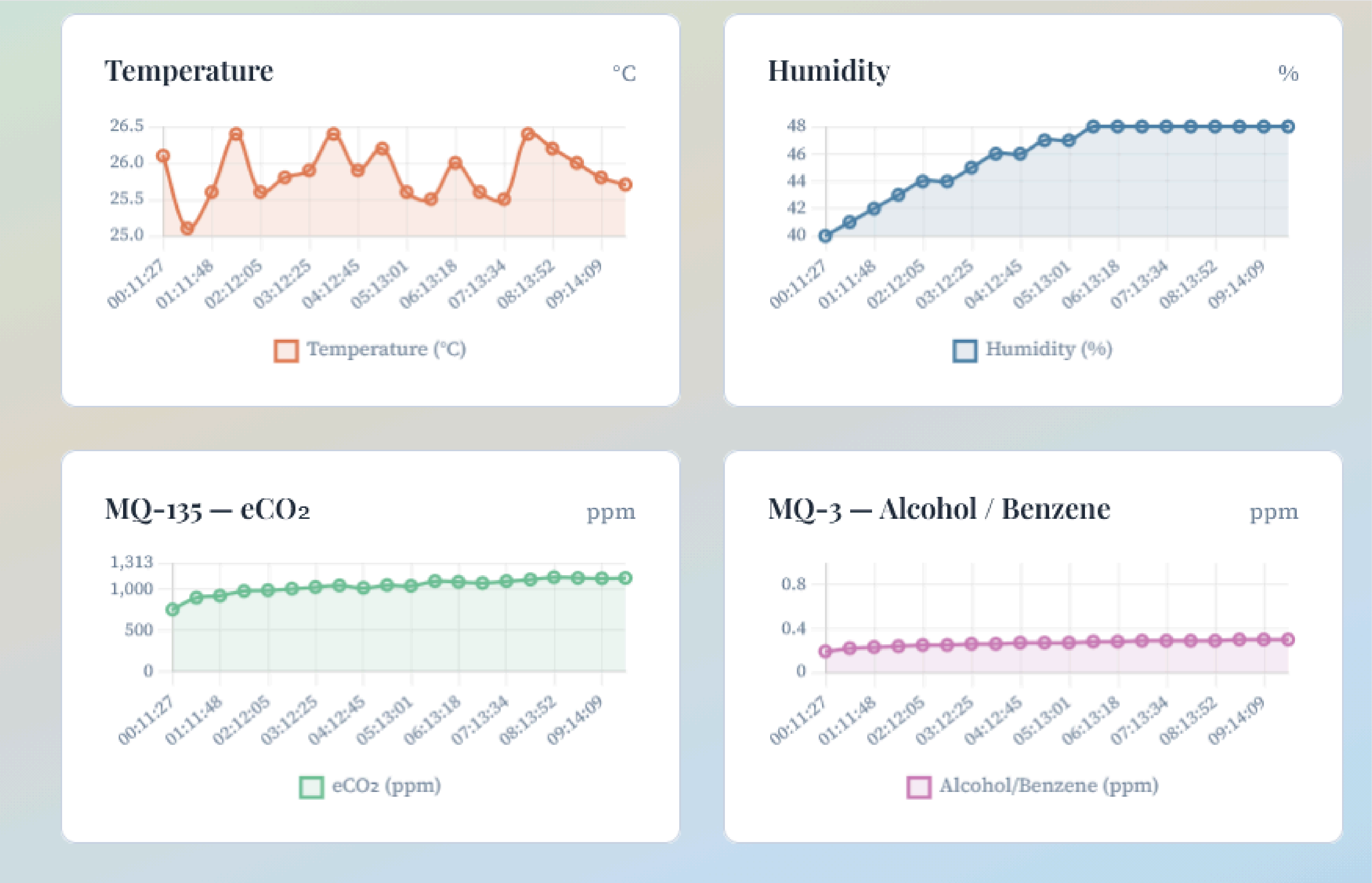
Sensor	May 25th PPM mean	June 13th PPM mean	Change
MQ135	267.81	1081.04	+304%
MQ3	0.58	0.28	-52%
MQ6	1299.27	1184.39	-9%
MQ7	150.46	98.95	-34%
MQ8	1152.35	1354.16	+17%

Value comparison between two different days (25th May before and 13th June after sensor calibration)

Comparison between values at the beggining and end of a session (June 13th)

Sensor	03:12 reading (ppm)	09:44 reading (ppm)	Variation
MQ3	0.25	0.3	+20%
MQ135	1022.61	1133.97	+10.9%
MQ6	1094.34	1238.04	+13.1%
MQ7	93.80	101.98	+8.7%
MQ8	1330.29	1375.58	+3.4%

RESULTS



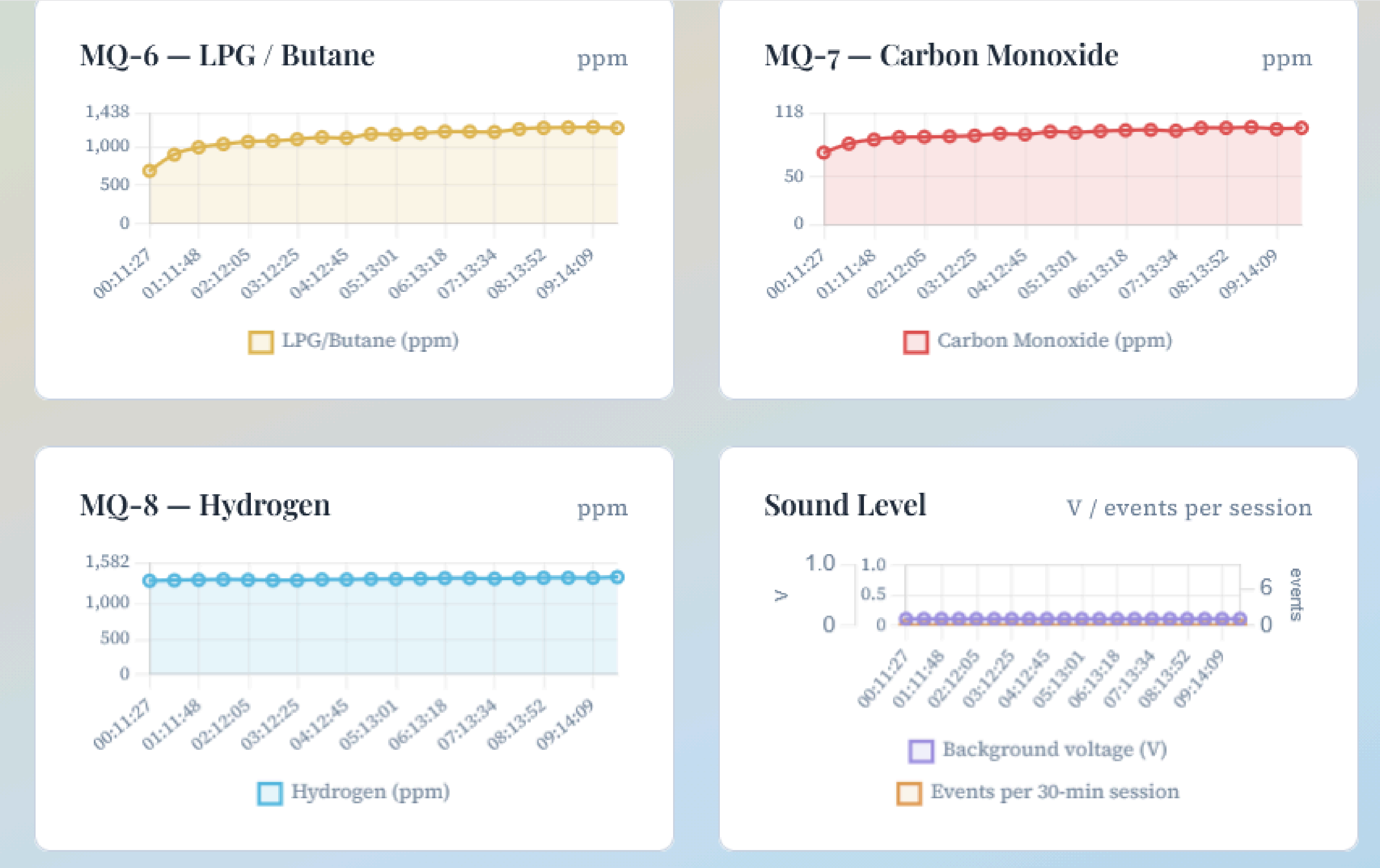
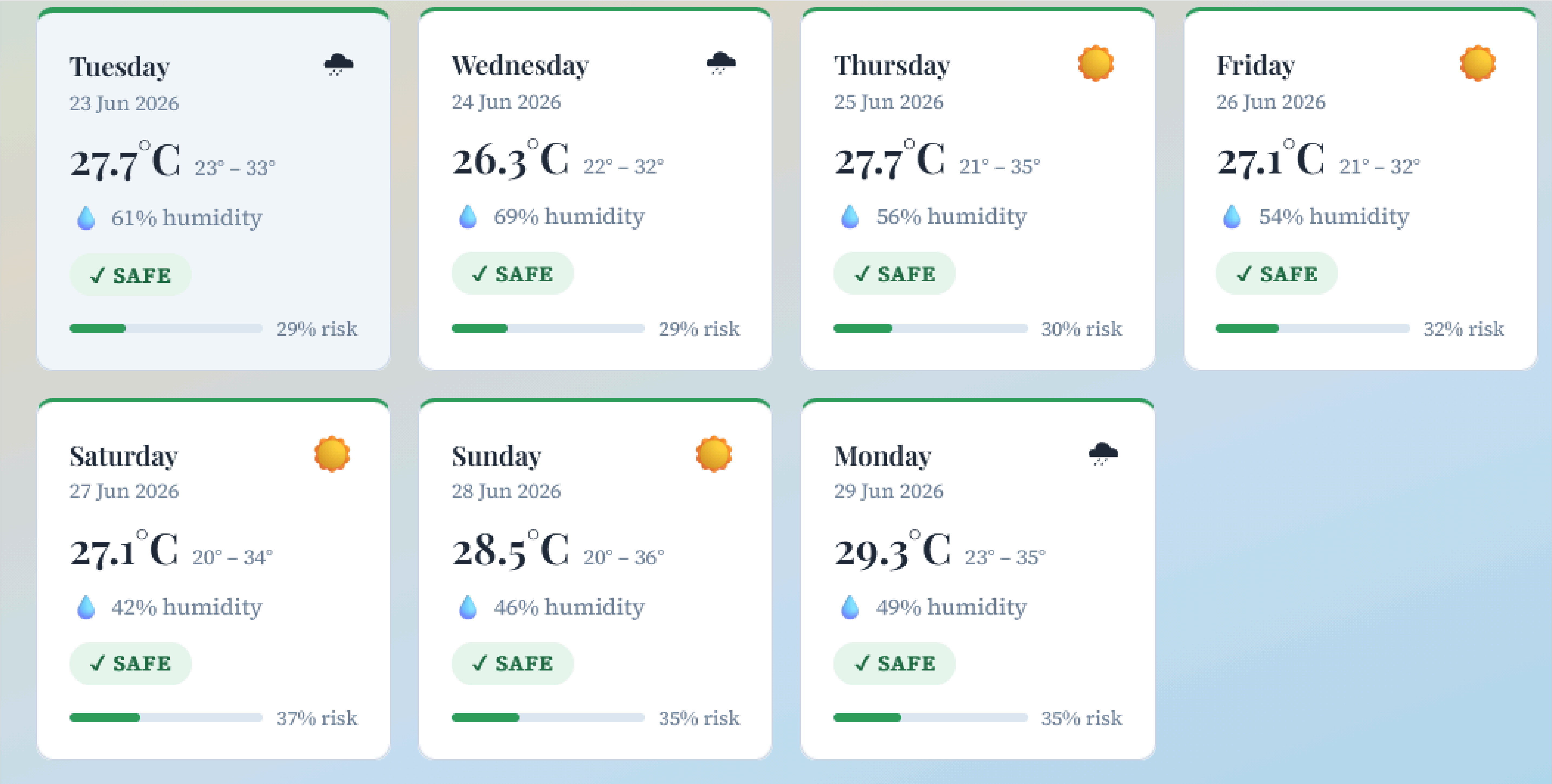


Chart data present on the pictures above is from June 13th

RESULTS



The predictions in the picture
were generated on 23.06.2026

CONCLUSION

This project presents an IoT-based smart indoor air quality monitoring product which integrates multiple sensors with a Raspberry Pi 4, cloud storage, web and mobile applications, and 2 machine learning prediction models. By combining all of these, the system offers a solution for improving indoor environmental awareness.

FUTURE WORK

- adding new sensors
- upgrading already existing sensors
- cooling option for the Raspberry Pi
- designing a protective case for the product
- further training of AI models for more accurate predicitons

THANK YOU!